

TABLE I
DIY-Type 3D Printers

Vendor	Product
3D Printronics	Felix 2.0
Lulzbot	TAZ 4
RS Components	RepRap Pro merod
Chipmax	MBot Cube
Amtronics	Prusa Mendel LMBUU

TABLE II
Assembled Desktop 3D Printers

Vendor	Product
3D Systems (formerly ZCorp)	Zprinter 850
Afinia	H479
3D Protomaker	Sprite
Makerbot	Replicator 2
Airwolf 3D	AW3D HD
Altam Technologies	uPrint SE
Amtronics Systems	Dual Extruder 3D Printer
Brahma 3	Brahma3 Anvil
CAD Centre	ProJet 7000 HD
Cubify	CubeX
Formlabs	Form 1
J Group Robotics	Dimension LE
KCBots	KUBE
Stratasys	Objet 1000

ovens, PCB-prototyping machines, stencil printers and 3D printers, all the above-mentioned prototyping work can now be done in-house and a lot of time and money can be saved. These equipments are expensive but, in the long run, can save you money and, most importantly, the time to market. Let us have a look at these equipments one by one.

3D printers

Using 3D printers, which got introduced commercially only recently, you can produce almost any part. These printers can be used for checking the mechanical design and fitment. In regular process, you get the designs made and then send them to the manufacturer. Then there is a long wait be-

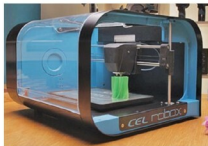


Fig. 1: A 3D printer

fore you receive the prototype to check everything and finalise the mechanics. With a 3D printer you can immediately fabricate your design and check it. It accepts .stl files and converts them to gcode files to print all layers one by one. Fig. 1 shows a 3D printer.

The material of the printed prototype will of course be different, but you only want to check the dimensions, fitment and integration of different parts. Sometimes your prototype could have dimensions larger than what your 3D printer can handle. In that case you can scale down the dimensions of each part and check the fitment and integration in this scaled down dummy.

3D printers also come in the form of DIY kits, such as Prusa Mendel I2. You get all the parts of the printer but you need to follow its manual to assemble it and install everything before you can produce anything. These printers are normally less precise and you have to do a lot of calibration before starting work on them. But such printers are usually cheaper and can be good for basic mechanical models' dummy checking. Table I lists some manufacturers of the DIY-type 3D printers.

But if you are looking for more precise and reliable prototyping, you should go for ready-to-use 3D printers. These printers come assembled with reliable mechanical structure and give precise results as per specs mentioned by their manufacturers. Most of all, nil or very little calibration is required in such printers. Table II shows some assembled desktop 3D printers from different manufacturers.

Here are some more pointers to

help you select the most suitable 3D printer for you:

Print platform. This is the maximum dimension that a printer can print. Select appropriate dimensions for your requirement. If the prototype is larger than that, either print a scaled down version or break it into parts that can later be combined.

Resolution. Check for horizontal and vertical resolutions in the specs sheet. The vertical resolution will define the thickness of each layer. Better resolution prints smoother surfaces. Similarly, horizontal resolution defines how fine the extruder can move in XY plane. The printer with finer XY resolution can print finer features in the model.

Print speed. This will give you an idea about how fast your designs can be printed. It is normally defined as the time it takes to print a specific distance in the Z-axis.

Number of extruders. Extruders are the print heads. The number of extruders is generally associated with the number colours that the printer can handle at a time.

SD card and display support. Some printers have an interactive display to help select all the functions, but this comes at an additional cost. SD card support is also an important feature; the design can be copied in the SD card and the printer keeps printing without the need of an attached computer.

Desktop PCB prototyping

With the mechanical design verified, the next step is to make your PCB design such that it fits into the mechanics. Once the design is done using PCB design software, within the dimensional constraints set by the mechanics, you will have to get some of these PCBs made so that you can mount all the components and test the boards for functionality.

Since most manufacturers are not interested in low quantities, it is best to make these PCBs in-house. Though some people use photo-resist and UV systems for making a few pieces, these techniques are inaccurate and time